

title: "Savitribai Phule Pune University --- Programming and Problem Solving"
code: "110005"
pattern: "2019"
total_marks: 70
duration: "2½ Hours"

Seat No.

SAVITRIBAI PHULE PUNE UNIVERSITY

Programming and Problem Solving (110005)

2019 Pattern | Semester I/II

Total Marks: 70 | Time: 2½ Hours

Instructions: Q1/Q2, Q3/Q4, Q5/Q6, Q7/Q8. Neat diagrams must be drawn wherever necessary.

Unit III --- Functions and Modules (18 Marks)

Q1) Attempt the following:

a) Define a function. Explain the need for functions in Python.

b) Differentiate between local and global variables with examples.

c) Write a program using a function to find the largest of 3 numbers by passing numbers as arguments.

OR

Q2) Attempt the following:

a) Explain the following types of function arguments with examples: i) **Default arguments**
ii) **Variable length arguments**

b) Write a program to find the square of a number using lambda function.

c) What are modules and packages in Python? Explain a user-defined module with example.

Unit IV --- Strings (17 Marks)

Q3) Attempt the following:

a) Explain different ways of formatting a string in Python with suitable examples.

b) Explain different ways of iterating a string.

c) What is the output of the following for string `S = "Learning Python is fun": i) print(S[:10])`
ii) `print(S[::-1])` iii) `print("Is" not in S)` iv) `print(S<span class="marks">[4]</span>)` v)
`print(S[0:10])`

OR

Q4) Attempt the following:

a) Explain the following string methods with examples: i) `isspace()` ii) `join()`
iii) `lower()`

b) Write a program to display a string and count the number of characters in the string using a loop.

c) Explain the `ord()` and `chr()` functions with examples. Explain `in` and `not in` operators on strings.

Unit V --- Object Oriented Programming (18 Marks)

Q5) Attempt the following:

a) Explain the following programming paradigms: i) **Monolithic Programming** ii) **Procedural Programming** iii) **Object Oriented Programming**

b) Explain the following OOP concepts with examples: i) **Class variable** ii) **Object variable**

c) Write a Python program to create a class `<code> Employee </code>` with attributes `<code> Name </code>`, `<code> emp_id </code>`, and `<code> salary </code>`

a) Explain the following features with examples: i) **Containership** ii) **Inheritance**
iii) **Data Abstraction**

b) Explain class method and self object with suitable example.

c) Write a program to create a class `Car` with two attributes `name` and `cost`. Create two objects and display their information.

Unit VI --- File Handling (17 Marks)

Q7) Attempt the following:

a) What is a file? Explain relative and absolute path of a file with examples.

b) Explain the following file handling methods: i) `read(n)` ii) `readlines()`
iii) `close()`

c) Explain the following file access modes: `r`, `w+`, `a+`, `rb`, `ab+`.

OR

Q8) Attempt the following:

a) Explain any three directory methods with examples.

b) Explain text and binary files with examples.

c) Explain any three dictionary methods with examples.

*** END OF QUESTION PAPER ***